

```
In [24]: # Step 1: Import required Libraries
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [4]: # Step 2: Load the dataset
df = pd.read_csv("C:\\Disk Partition\\Jupyter\\car_parts_sales.csv")
```

```
In [5]: df.head()
```

Out[5]:

	order_id	order_date	customer_id	customer_name	product_category	product_name	product_price	quantity	total_amount	payment_method
0	10001	2026-05-03	2410	Kyle Bishop	Engine Parts	Oil Filter	2482.20	7	17375.40	Card
1	10002	2026-03-30	9234	Ricky Scott	Engine Parts	Timing Belt	3363.12	1	3363.12	Card
2	10003	2026-04-25	1342	Donald Sosa	Interior	Seat Cover	1436.42	7	10054.94	COD
3	10004	2025-12-11	4417	Eric Bradford	Interior	Floor Mat	3276.09	10	32760.90	COD
4	10005	2026-08-08	3207	Adam Hall	Suspension	Coil Spring	2789.51	7	19526.57	Wallet

```
In [29]: # Summary statistics for numerical columns
df.describe()
```

Out[29]:

	order_id	customer_id	product_price	quantity	total_amount
count	100.000000	100.000000	100.000000	100.000000	100.000000
mean	10050.500000	5490.960000	2912.279500	5.5200	15825.992200
std	29.011492	2557.23316	1293.199161	2.6911	11120.900716
min	10001.000000	1055.000000	625.260000	1.0000	1992.390000
25%	10025.750000	3451.000000	1780.497500	3.0000	6532.420000
50%	10050.500000	5442.000000	3000.370000	5.5000	13306.760000
75%	10075.250000	7583.500000	4031.905000	8.0000	22798.752500
max	10100.000000	9838.000000	4993.850000	10.0000	48744.600000

```
In [30]: # Check structure of dataset
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   order_id              100 non-null   int64
1   order_date            100 non-null   str
2   customer_id           100 non-null   int64
3   customer_name         100 non-null   str
4   product_category      100 non-null   str
5   product_name          100 non-null   str
6   product_price         100 non-null   float64
7   quantity              100 non-null   int64
8   total_amount          100 non-null   float64
9   payment_method        100 non-null   str
dtypes: float64(2), int64(3), str(5)
memory usage: 7.9 KB
```

```
In [11]: # Count unique values in product category column
df['product_category'].value_counts()
```

Out[11]:

product_category	
Electrical	27
Engine Parts	25
Suspension	22
Interior	17
Brake System	9
Name: count, dtype: int64	

```
In [12]: # Count unique values in payment method column
df['payment_method'].value_counts()
```

Out[12]:

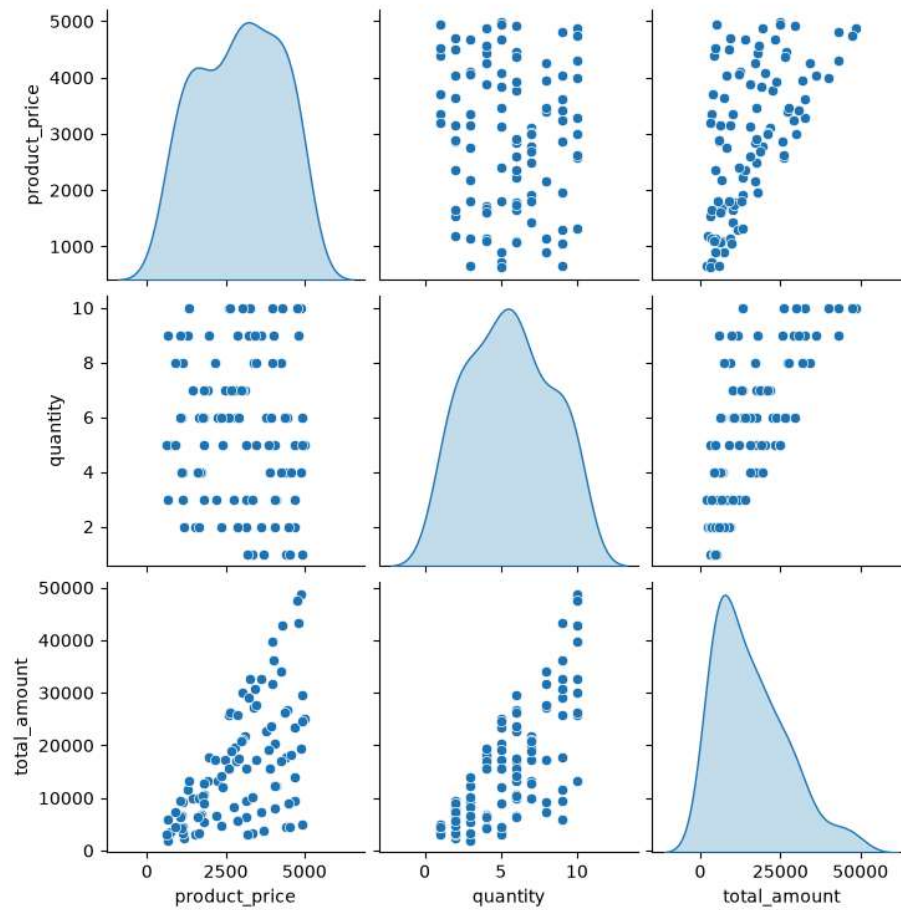
payment_method	
Wallet	27
UPI	26
Card	25
COD	22
Name: count, dtype: int64	

Observation 1

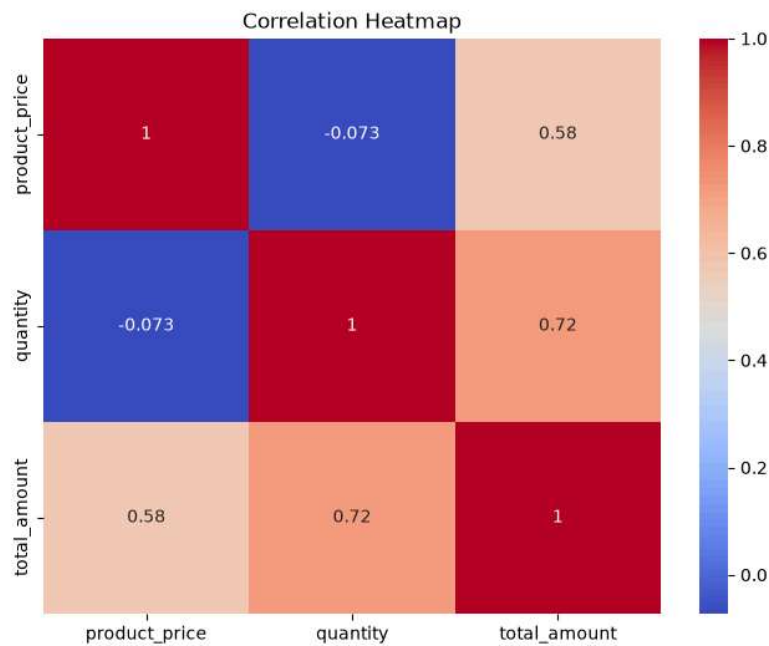
Correlations: Strong positive correlation between quantity and total_amount (expected), moderate correlation between product_price and total_amount

```
In [36]: # Pairplot for numerical features
sns.pairplot(df[['product_price', 'quantity', 'total_amount']], diag_kind='kde')
plt.suptitle("Pairplot of Numerical Features", y=1.08)
plt.show()
```

Pairplot of Numerical Features



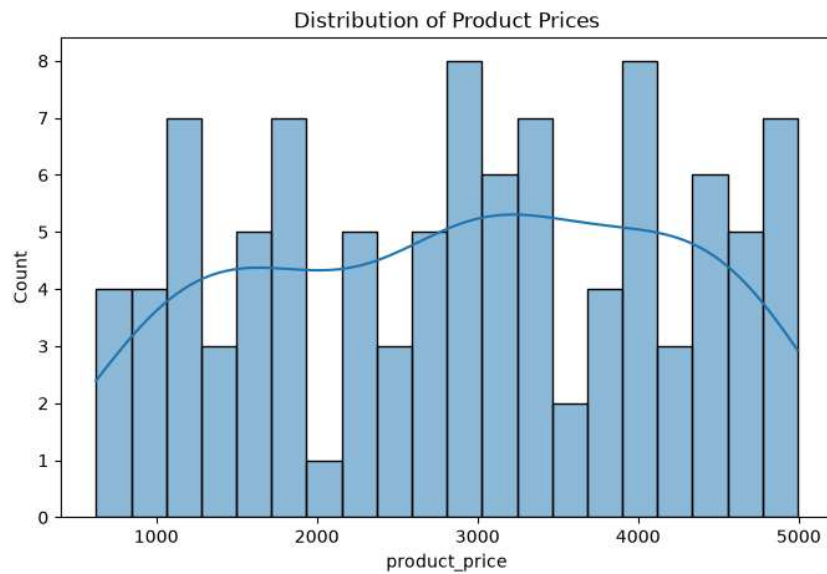
```
In [31]: # Correlation heatmap
plt.figure(figsize=(8,6))
sns.heatmap(df[['product_price', 'quantity', 'total_amount']].corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



Observation 2:

Price Distribution: Most products are priced between ₹1,000–₹5,000, with a few values significantly higher than others.

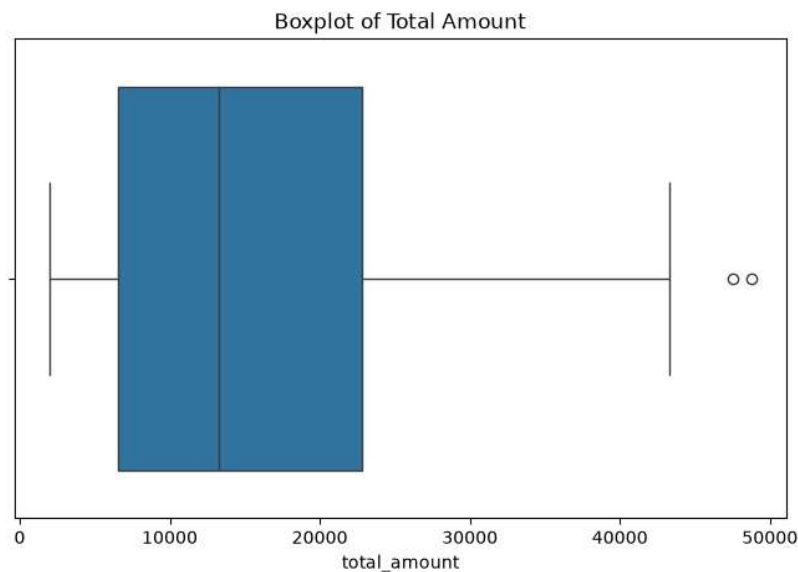
```
In [37]: # Histogram of product prices
plt.figure(figsize=(8,5))
sns.histplot(df['product_price'], bins=20, kde=True)
plt.title("Distribution of Product Prices")
plt.show()
```



Observation 3

Total Amount: Some very large transactions (₹40k+) suggest bulk purchases

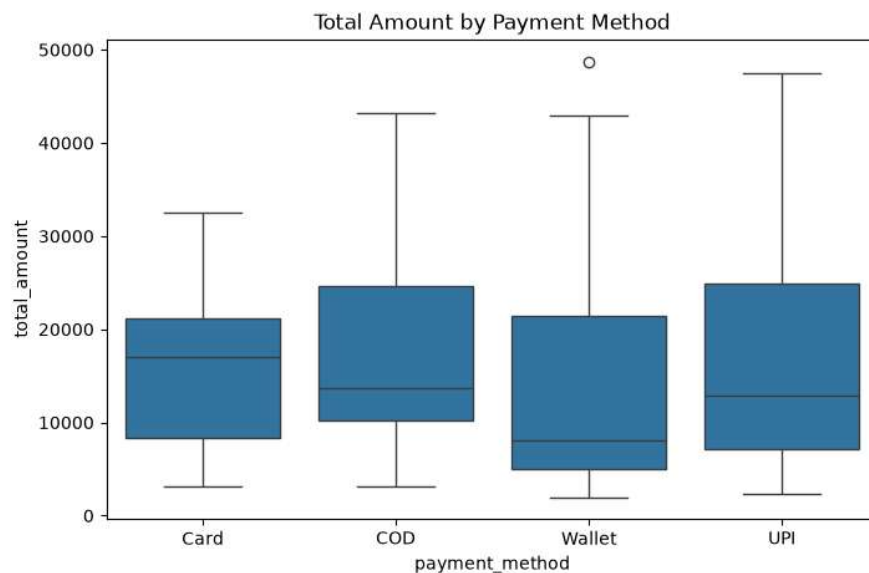
```
In [41]: # Boxplot of total amount
plt.figure(figsize=(8,5))
sns.boxplot(x=df['total_amount'])
plt.title("Boxplot of Total Amount")
plt.show()
```



Observation 4

Payment Methods: Wallet and COD are common, but high-value purchases lean toward Card/UPI.

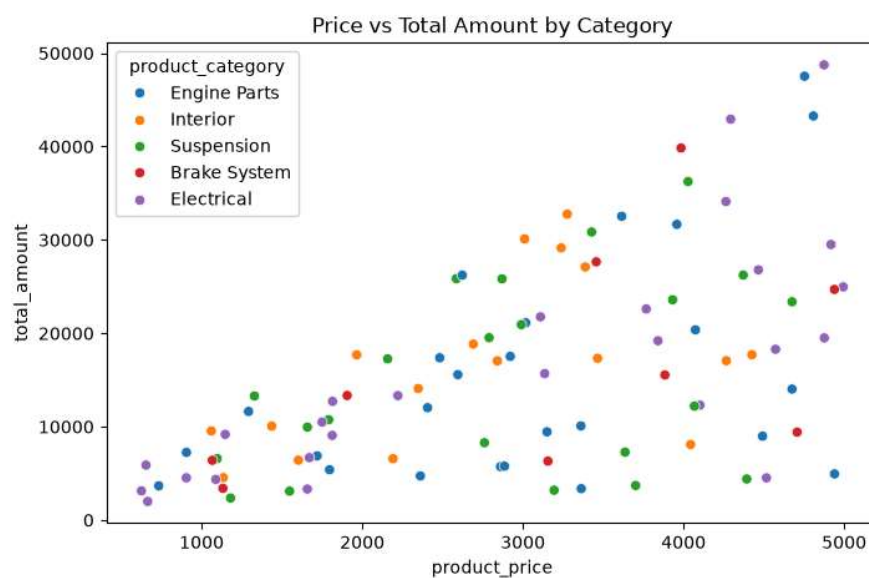
```
In [38]: # Boxplot: total amount by payment method
plt.figure(figsize=(8,5))
sns.boxplot(x='payment_method', y='total_amount', data=df)
plt.title("Total Amount by Payment Method")
plt.show()
```



Observation 5

Category Trends: Suspension and Electrical parts often have higher transaction totals.

```
In [39]: # Scatterplot: product price vs total amount
plt.figure(figsize=(8,5))
sns.scatterplot(x='product_price', y='total_amount', data=df, hue='product_category')
plt.title("Price vs Total Amount by Category")
plt.show()
```



8. Summary of Findings

- **Top Categories:** Electrical and Engine parts dominate sales volume with 50.5% of total sales.
- **High-Value Customers:** Large purchases are concentrated in Brake System and Engine Parts.
- **Payment Insights:** Wallet is popular for mid-range purchases, while Card/UPI are used for higher amounts.
- **Outliers:** A few extreme transactions may need further investigation (possible bulk orders or anomalies).